

Binary Adjacency Range Encoding (BARE)

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Abstract

Bare is a binary file format for storing **Graph Adjacency Lists** (GALs) as **Directly Addressable Byte Ranges** (DBRs) a.k.a *Dabbers*. It is based on **Binary Adjacency Range** (BAR), which is a storage model around a simple idea.

A vertex's adjacency list should be locatable as a contiguous Byte range.

Bare uses this approach to explore storing graph data via **Adjacency Lists** at scale. This paper contains details on the motivations and design choices made as well as additional implementation details.

Introduction

BARE uses binary encoded data written to disk rather than relying on native data structures *e.g. Dictionaries, Vectors etc.*

It essentially acts as a physical graph representation that maps a vertex's adjacency list to a directly addressable byte range. *See: CSR, WebGraph, BVGraph, Ligra, and GraphChi*